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Educating pharmacists for an artificial intelligence augmented profession: A narrative review of curricular and accreditation imperatives

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ABSTRACT

Generative artificial intelligence (AI) moved from novelty to clinical infrastructure within a single Doctor of Pharmacy program, yet pharmacy curricula and the accreditation standards that govern them have not uniformly caught up. Written from a Canadian vantage point, this narrative review synthesizes international peer-reviewed literature and grey-literature guidance from the International Pharmaceutical Federation, World Health Organization, UNESCO, and national accreditors in Canada, the United States, the United Kingdom, Australia, and New Zealand. Rather than presenting a flat catalogue of competencies, we propose a four-tier hierarchy that makes explicit which AI competencies pharmacy owns and which it shares: general competencies (foundational AI literacy and the applied skills that follow from it, which in steady state belong upstream of pharmacy education); interprofessional competencies (AI governance, ethics, equity, and regulatory awareness, shared across the health professions); pharmacy-specific competencies (clinical AI applications in pharmacy workflows, pharmaceutical sciences AI, and AI-augmented pharmacokinetics and model-informed precision dosing); and role-specific specialization. Because current students entered pharmacy programmes without K–12 or undergraduate AI preparation, faculties must carry the general and interprofessional layers transitionally, a five to ten year window in which faculty development, not curriculum design, is the binding constraint. Six pedagogical principles and five accreditation levers are synthesized, with CCAPP Standard 20 and its international counterparts named as the single highest-leverage accreditation instrument. Whether pharmacy remains the medication-therapy expert profession depends on whether graduates are the clinicians most literate in AI assisted pharmacotherapy and, most distinctively, in model-informed precision dosing, where the profession's claim is strongest.

1. Introduction

The typical Doctor of Pharmacy (PharmD) program spans four years. A student who began in 2023 entered pharmacy school without ever having used a large language model in coursework; that same student will graduate in 2027 into a practice environment in which AI-enabled clinical decision support systems recommend doses, ambient AI scribes draft progress notes, model-informed precision dosing (MIPD) platforms are deployed at more than a thousand hospital sites internationally, drug discovery pipelines are being reshaped by generative molecular design, and national regulatory frameworks for machine-learning-enabled medical devices (MLMDs) are now operative in Canada and other jurisdictions.[1–3] The curriculum that trained those students did not anticipate this transition. In most jurisdictions the accreditation standards governing that curriculum remain silent on it. The gap is neither hypothetical nor small.

The International Pharmaceutical Federation (FIP) has identified digital health and digital tools as priorities within its Development Goals for the pharmacy and pharmaceutical sciences workforce, acknowledging that the profession's capacity to adopt and shape digital innovation will determine its relevance across the coming decade.[4,5] The World Health Organization (WHO) has published successive guidance on the ethics and governance of AI for health, including the specific challenges posed by large multi-modal models.[6,7] UNESCO has issued guidance for generative AI in education and research,[8] and the European Union has enacted a comprehensive AI regulation.[9] Within pharmacy

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education specifically, commentators have argued that the pharmacy academy must be at the centre of discussions addressing the technical, philosophical, and ethical issues that AI presents for the future of the profession,[10] and more recent commentary has framed the training of pharmacy students in the use of AI for clinical practice as a professional obligation rather than an optional enrichment.[11] Yet translation of these supranational signals and academic calls into pharmacy-specific curricular and accreditation requirements has been uneven.

This narrative review synthesizes the international literature on AI in pharmacy education with three aims: first, to characterize what pharmacy education frameworks currently include and lack with respect to AI; second, to describe a set of AI competency domains that the contemporary literature supports as essential at entry to practice, organized as a hierarchy that distinguishes what pharmacy owns from what it shares; and third, to summarize pedagogical and accreditation levers available to programmes and regulators, with faculty development treated as the binding constraint. A secondary argument runs alongside the hierarchy: model-informed precision dosing is the single domain in which pharmacy's claim on AI-augmented practice is least contested and most cleanly defended, and we treat it accordingly as the anchor of the pharmacy-specific tier. The review is written from a Canadian vantage point and uses Canadian accreditation and educational frameworks as its central case. International best practices from the United States, the United Kingdom, Australia, New Zealand, and supranational bodies including FIP, WHO, and UNESCO are examined comparatively to identify transferable lessons for the Canadian context and, reciprocally, to surface Canadian precedents that may be instructive for peer jurisdictions.

2. Approach to the narrative review

This is a narrative rather than a systematic review, and it is not registered. A narrative approach was selected because the objective was framework development and policy synthesis rather than comprehensive mapping of all educational interventions. Sources were identified through iterative searches of PubMed, Embase, ERIC, and Google Scholar between 2019 and 2026 using combinations of the terms "artificial intelligence", "machine learning", "large language model", "pharmacy education", "pharmacy curriculum", "accreditation", "model-informed precision dosing", "clinical decision support", and "health professions education". Grey literature was drawn from FIP, WHO, UNESCO, Health Canada, the United States Food and Drug Administration (FDA), the United Kingdom Medicines and Healthcare products Regulatory Agency (MHRA), the Accreditation Council for Pharmacy Education (ACPE), the Canadian Council for Accreditation of Pharmacy Programs (CCAPP), the Association of Faculties of Pharmacy of Canada (AFPC), the General Pharmaceutical Council (GPhC) of Great Britain, the Australian Pharmacy Council (APC), the Pharmacy Council of New Zealand (PCNZ), and the European Commission. Priority was given to documents and studies published from 2021 onward to reflect the rapid transformation of the field following the public release of generative AI; earlier landmark work was retained where conceptually foundational. The regulatory and accreditation snapshot described below reflects documents publicly available up to early 2026; given the pace of change in this domain, readers are encouraged to verify current versions of any standard or guidance at the time of application. The synthesis is interpretive rather than quantitative and is intended to characterize the field rather than to estimate a summary effect.

3. Findings: global policy context

Pharmacy education operates within an increasingly dense AI policy environment. The FIP Development Goals articulate a workforce transformation agenda in which digital fluency is foundational,[4,5] and FIP has since addressed the integration of digital health content into pharmacy curricula.[12] At the health-system level, WHO has framed ethical and governance expectations for AI applications in health covering transparency, accountability, equity, and redress and has issued specific guidance addressing the risks of large multi-modal models deployed in clinical workflows.[6,7] UNESCO guidance on generative AI in education and research articulates the responsibilities of educators and institutions in managing student use of generative tools while supporting equitable access and academic integrity.[8] Professional pharmacy organizations were addressing artificial intelligence even before generative AI's mainstream adoption: ASHP's 2020 statement on the use of AI in pharmacy anticipated many of the workforce and workflow questions since raised by generative tools,[13] and FIP's global survey of pharmacy schools found preparedness and responsiveness to digital-health training highly uneven across member countries, a baseline against which subsequent progress can be measured.[14]

The regulatory frameworks governing the AI tools pharmacists use have moved materially in the last twenty-four months. Health Canada's February 2025 Pre-market Guidance for Machine-Learning-Enabled Medical Devices introduced a formal framework for MLMD classification, pre-determined change control plans, transparency requirements, and post-market performance monitoring.[1] The harmonized Good Machine Learning Practice guiding principles jointly issued by Health Canada, the FDA, and the MHRA explicitly include "focus on the performance of the human AI team" as a core principle, and expect that users receive clear, essential information about model function.[15,16] The European Union AI Act establishes a risk-based regulatory regime that designates many health-related AI systems as high-risk, imposing transparency, data quality, and human oversight obligations.[9] These are not internal industry documents; they are the lens through which regulators assess the devices pharmacists use, and the ecosystem within which graduates will practice. Table I summarizes the key regulatory and policy instruments relevant to pharmacy practice and education from 2021 to 2025.

4. Findings: current state of AI in pharmacy curricula and accreditation

Most national pharmacy education frameworks were designed before generative AI and include limited or no explicit AI content. In Canada the framework comprises three interlocking documents: the AFPC Educational Outcomes for First Professional Degree Programs in Pharmacy in Canada 2017, which define seven pharmacist roles (Care Provider, Communicator, Collaborator, Leader-Manager, Health Advocate, Scholar, and Professional)[17]; the NAPRA Professional Competencies for Canadian Pharmacists at Entry to Practice, which specify what a licensed pharmacist must be able to do on day one.[18] and CCAPP's Accreditation Standards,[19] which require every accredited PharmD program to base its curriculum on both. An AFPC-Canada Health Infoway partnership has developed thirty-one Pharmacy Informatics Entry-to-Practice Competency Indicators across three domains: A. information and knowledge management; B. professional and regulatory accountability; and C. information and communication technologies, supported by open-access e-learning modules.[20] The foundation is not absent; it is, however, calibrated to the informatics world of approximately 2017. The CCAPP 2025 standards, which comprise twenty-five standards across three parts, contain no explicit reference to artificial intelligence, machine learning, algorithmic bias, or model governance. Standard 25 addresses "information resources" and educational technology but does not extend to AI-specific competencies.

In the United States, ACPE has issued guidance on AI use in continuing pharmacy education, and the American Association of Colleges of Pharmacy (AACP) convenes a recurring AI Education Institute aimed at faculty capacity.[21,22] The updated ACPE Standards 2025, effective 1 July 2025, retain health informatics as a required curricular element but do not name artificial intelligence explicitly.[23] In the United Kingdom, the General Pharmaceutical Council's Standards for the Initial Education and Training of Pharmacists require pharmacy graduates to

Table 1
Selected regulatory and policy instruments relevant to pharmacy practice and education, 2021–2025.

Year	Instrument	Issuing body	Relevance to pharmacy education
2021	Good Machine Learning Practice for Medical Device Development: Guiding Principles	Health Canada, FDA, MHRA	Ten principles including explicit focus on human–AI team performance; frames expectations of transparency and user information.
2021	Ethics and Governance of Artificial Intelligence for Health: WHO Guidance	World Health Organization	Articulates transparency, accountability, equity, and redress expectations applicable to health AI.
2021	Standards for the Initial Education and Training of Pharmacists	General Pharmaceutical Council (Great Britain)	Current UK pharmacist education standards; digital/informatics skills required but AI not specified.
2022	Accreditation Standards for Pharmacy Programs 2020 (updated October 2022)	Australian Pharmacy Council	Current Australian accreditation standards; embed Aboriginal and Torres Strait Islander cultural safety but do not name AI.
2023	Guidance for Generative AI in Education and Research	UNESCO	Guidance on managing student use of generative AI in education and research.
2024	Transparency for Machine Learning-Enabled Medical Devices: Guiding Principles	Health Canada, FDA, MHRA	Extends GMLP principles specifically to transparency expectations for MLMD users.
2024	Regulation (EU) 2024/1689 (Artificial Intelligence Act)	European Parliament and Council	Risk-based regulatory regime designating many health-related AI systems as high-risk.
2024	Accreditation Standards for Pharmacy Programmes; Competence Standards for the Pharmacy Profession (both effective 1 April 2024)	Pharmacy Council of New Zealand	Te Tiriti o Waitangi as dedicated competence-standard domain; AI not named.
2024	Ethics and Governance of Artificial Intelligence for Health: Guidance on Large Multi-Modal Models	World Health Organization	Extends 2021 WHO guidance to the specific risks of LMMs in clinical workflows.
2025	Pre-market Guidance for Machine-Learning-Enabled Medical Devices (February 2025)	Health Canada	Introduces MLMD classification, predetermined change control plans, and post-market performance monitoring in Canada.
2025	Accreditation Standards for Canadian Educational Programs Leading to the Doctor of Pharmacy (Pharm.D.) Degree (April 2025 revision)	Canadian Council for Accreditation of Pharmacy Programs	Current Canadian PharmD accreditation standards; AI not explicitly named.
2025	Standards 2025: Accreditation Standards and Key Elements for the Professional Program in Pharmacy (effective 1 July 2025)	Accreditation Council for Pharmacy Education	Current US pharmacy accreditation standards; health informatics required but AI not explicitly named.

Note. FDA = U.S. Food and Drug Administration; GMLP = Good Machine Learning Practice; LMM = large multi-modal model; MHRA = U.K. Medicines and Healthcare products Regulatory Agency; MLMD = machine-learning-enabled medical device.

demonstrate professional knowledge and skills applicable to digital and informatics contexts, though AI-specific content is not explicitly mandated at the learning outcome level.[24] Early pedagogical interventions (for example, a multi-institutional study using structured case-based learning and prompt-engineering frameworks in pharmacotherapy courses) illustrate the kind of localized innovation now appearing in response,[25] and recent scoping reviews of AI and digital health innovations in pharmacy education confirm that implementation across curricula remains limited and uneven.[26,27] A 2026 scoping review of AI integration and global perceptions across pharmacy education corroborates this picture, reporting that although faculty and students show high willingness to adopt AI, integration remains inconsistent and AI literacy highly variable.[28] The broader digital-health literature anticipated this unevenness: writing before the generative-AI era, Aungst and Patel argued that health-professions curricula were already lagging behind the pace of digital-health innovation, a gap that the subsequent AI transition has widened rather than closed.[29]

The Australian and New Zealand experience is particularly instructive for Canadian observers. In Australia, the APC accredits pharmacy degree and intern training programmes against the Accreditation Standards for Pharmacy Programs 2020 (updated October 2022), with graduates then approved for provisional registration by the Pharmacy Board of Australia under the National Registration and Accreditation Scheme.[30] These standards were historically applied jointly to Australia and New Zealand until PCNZ transitioned to in-house accreditation: PCNZ published its own accreditation and competence standards in June 2023 and brought all new standards into effect on 1 April 2024, partly to better align with local priorities such as Te Tiriti o Waitangi.[31,32] Neither framework yet names artificial intelligence, machine learning, or algorithmic governance as explicit competency requirements, a pattern that parallels CCAPP's current standards and is similarly evident in ACPE's Standards 2025.[23] What the Australian and New Zealand frameworks do offer, however, is a directly relevant structural precedent: the integration of Aboriginal and Torres Strait Islander health and cultural safety in Australia, and of Te Tiriti o Waitangi as a dedicated competence-standard domain in New Zealand,[32] demonstrates how national accreditors can mandate new, transversal competencies through accreditation architecture rather than waiting for bottom-up curricular adoption. This is the same mechanism that Canadian pharmacy used in response to the Truth and Reconciliation Commission.[33] and that could plausibly be applied to AI competence in the next CCAPP revision cycle.

The common gap is that accreditation standards in most jurisdictions describe the shape of the curriculum without requiring the AI specific competencies that practice now demands. A particularly telling instance of that silence is MIPD: the pharmacy-proximal AI application with the most mature evidence base, and the domain in which pharmacy has the sharpest distinctive claim, is not explicitly required anywhere in the five jurisdictions surveyed here. Standards-setting cycles are slower than the technology adoption cycle; an accreditation revision that responds to AI only in the next ten-year review will deliver AI-competent graduates in the late 2030s. Table II summarizes current accreditation standards across the five jurisdictions surveyed here, highlighting the consistent absence of explicit AI content alongside the presence of transversal cultural-competence precedents that offer structural models for future AI integration.

5. Findings: Pedagogical evidence from the literature

Beyond policy and accreditation documents, a growing empirical literature reports directly on how AI is being taught, and how faculty and students are responding. A 2026 analysis of educational strategies and implementation challenges at a US college of pharmacy reported

Table 2
Cross-jurisdictional comparison of current pharmacy accreditation frameworks with respect to artificial intelligence (AI) content.

Jurisdiction	Current standards (accreditor; effective date)	AI explicit?	Digital/ informatics coverage	Related cultural-competence precedent
Canada	PharmD Accreditation Standards (CCAPP; April 2025)	No	Standard 25 (information resources, educational technology); AFPC–Infoway Pharmacy Informatics Entry-to-Practice Competencies	Indigenous health integration via Truth and Reconciliation Commission Calls to Action
United States	Standards 2025 (ACPE; effective 1 July 2025)	No	Health informatics named as a required curricular element; ACPE guidance issued for AI use in continuing pharmacy education	Cultural humility and health-equity requirements within standards
United Kingdom	Standards for the Initial Education and Training of Pharmacists (GPhC; 2021)	No	Digital and informatics professional knowledge and skills required, without AI-specific learning outcomes	Person-centred-care and equality, diversity, and inclusion standards
Australia	Accreditation Standards for Pharmacy Programs 2020 (APC; updated October 2022)	No	Implicit in program standards; no AI-specific criterion	Aboriginal and Torres Strait Islander cultural safety embedded across the standards
New Zealand	Accreditation Standards for Pharmacy Programmes and Competence Standards (PCNZ; both effective 1 April 2024)	No	Implicit in program standards; no AI-specific criterion	Te Tiriti o Waitangi as dedicated Domain 1 of the Competence Standards

Note. ACPE = Accreditation Council for Pharmacy Education (United States); AFPC = Association of Faculties of Pharmacy of Canada; APC = Australian Pharmacy Council; CCAPP = Canadian Council for Accreditation of Pharmacy Programs; GPhC = General Pharmaceutical Council (Great Britain); PCNZ = Pharmacy Council of New Zealand.

that faculty AI familiarity and student preparedness both lag curricular ambition and that integration, rather than a stand-alone course, is the practical path forward.[34] A national survey of pharmacy faculty and students similarly found that, although most respondents reported at least some familiarity with AI tools, faculty were less likely than students to trust AI-generated information despite comparable rates of encountering incorrect output, and faculty had sought formal training more often than students, pointing to a persistent readiness gap between generations of the profession.[35] Multi-programme pilots are beginning to translate this evidence into practice: a five-school collaboration using large-language-model-simulated patients to teach self-care patient-interviewing skills found the approach easy to use and well received by students, with most students supporting its continued use in the curriculum, illustrating how AI can be embedded directly into existing skills laboratories rather than taught as a separate subject.[36] Table III maps these and the other empirical sources synthesized in this review to their setting, AI domain, and principal finding; the implementation and policy recommendations we draw from this evidence are presented separately, in Sections 8 through 11.

6. A proposed hierarchy of AI competencies for pharmacy practice

The literature supports a range of AI competencies now relevant to pharmacy, but a flat list has two weaknesses. It reads as a catalogue rather than an argument, and it does not answer the accountability question that accreditors must ultimately rule on: who is responsible for producing each competency? Building on the findings summarized in Sections 3 through 5, we organize the competencies into four tiers, moving from most general to most pharmacy-specific, and, within the general and interprofessional tiers, distinguish "literacy" (knowledge and comprehension) from "skill" (application). Table IV summarizes the hierarchy and maps each competency to AFPC pharmacist roles and example pedagogical strategies supported by the reviewed literature. Fig. 1 depicts the hierarchy schematically, distinguishing the tiers that pharmacy education owns in steady state from those it carries only transitionally.

6.1. General competencies: foundational AI literacy and the applied skills that follow

Foundational AI literacy is the knowledge base required to make informed decisions about AI tool selection and use. At minimum, graduates should be able to distinguish among different families of AI approach (for example, classical statistical inference versus machine-learning prediction; narrow, task-specific AI versus general-purpose foundation models; supervised, unsupervised, and reinforcement learning), to recognize the concepts that govern model behaviour (training data, validation, overfitting, generalization, distributional drift, hallucination), and to understand the limitations and failure modes those concepts imply.[37,38] The purpose of this literacy is not technical fluency for its own sake; it is the decisional competence to choose when to use, when to trust, and when to refuse an AI tool.

From that knowledge base, two applied skills follow. The distinction is deliberate and Bloom-style: literacy is knowledge and comprehension; skill is application. The first is critical appraisal of AI outputs. The AFPC Scholar role and its international analogues already require critical evaluation of drug information sources; that competency must be extended explicitly to AI outputs. Graduates should cross-check an AI-generated drug recommendation against primary literature; recognize confabulated citations, which remain a persistent failure mode of generative models; assess whether a model's training and validation populations are representative of the patient in front of them; and interrogate the uncertainty of an AI prediction.[39] The second skill is effective human AI teaming, which encompasses structured prompting (including scaffolds such as CLEAR[39]), recognition of automation bias and alert fatigue, calibration of trust to the demonstrated reliability of the tool at hand, and knowing when to override. The Good Machine Learning Practice guiding principles explicitly frame "focus on the performance of the human AI team" as a core principle alongside expectations of clear, essential information to users.[15] Literacy without skill produces graduates who can describe AI but not use it; skill without literacy produces graduates who use AI without understanding when it will fail.

In steady state, neither the literacy nor the skills in this tier should sit inside the pharmacy curriculum proper. They belong upstream: in K-12 digital-literacy curricula, in undergraduate pre-pharmacy science, statistics, and research-methods courses, and in institution-wide digital-literacy infrastructure. The transitional problem this creates is discussed explicitly in Section 7; its implication for accreditation is that faculty

Table 3
Summary of empirical and pedagogical literature synthesized in this review, mapped to setting and principal finding, in reference-number order.

Ref.	Source	Setting	AI domain	Principal finding
2	Joshi & Sheth, 2025	Narrative review	Pharmaceutical sciences	AI reshapes formulation design and dosage-calculation workflows across the drug development pipeline.
3	Poweleit et al., 2023	Narrative review	Model-informed precision dosing	AI/ML approaches increasingly support therapeutic drug monitoring and individualized Bayesian dose forecasting.
10	Cain et al., 2023	Perspective, USA	Curricular strategy	Argues pharmacy education must define AI's role deliberately rather than respond ad hoc.
11	Nagel et al., 2026	Commentary, USA	Curricular obligation	Argues pharmacy programmes are obligated to train students in clinical AI use; identifies faculty readiness as the binding constraint.
13	Schutz et al., 2020	Policy statement, USA	Professional policy	ASHP statement anticipating workforce and workflow questions later raised by generative AI.
14	Mantel-Teeuwisse et al., 2021	Global survey (FIP)	Digital health education	Preparedness and responsiveness of pharmacy schools to digital-health training is highly uneven across member countries.
25	Munusamy et al., 2026	Intervention study, USA	Pharmacology teaching	Structured case-based learning and prompt-engineering frameworks embedded in pharmacotherapy courses promoted measurable AI readiness.
26	Abdel Aziz et al., 2024	Scoping review, USA	Curricular integration	Confirms AI implementation across pharmacy curricula remains limited and uneven.
27	Alsulami, 2025	Scoping review, global	Digital health integration	Corroborates uneven integration of digital-health innovations, including AI, across programmes.
28	Kattan et al., 2026	Scoping review, global	Curricular integration and perceptions	High willingness to adopt AI among faculty and students, but integration inconsistent and AI literacy highly variable.
29	Aungst & Patel, 2020	Perspective, USA	Digital health curricula	Health-professions curricula already lagged the pace of digital-health innovation before the generative-AI era.
34	Nguyen et al., 2026	Educational strategy analysis, USA	Curricular integration	Faculty AI familiarity and student preparedness both lag ambition; integration, not a stand-alone course, is the practical path forward.
35	Gustafson et al., 2025	National survey, USA	Faculty/student readiness	Faculty trusted AI-generated information less than students despite comparable error rates, and had received more formal training.
36	Walter et al., 2026	Multi-site implementation study, USA (5 programmes)	Skills-laboratory teaching	LLM-simulated patients for interviewing-skills practice were easy to use, scalable, and well received by students.
37	Rajkomar et al., 2019	Narrative review	Foundational AI literacy	Establishes core machine-learning concepts (training, validation, generalization) relevant to clinical AI literacy.
38	Topol, 2019	Monograph	Foundational AI literacy	Foundational text on AI's implications for clinical practice and the human-AI relationship.
39	Lo, 2023	Framework paper	Critical appraisal/ prompting	Introduces the CLEAR prompt-engineering framework for structured, critically appraised AI use.
40	Amaral Silva et al., 2022	Case report, Canada	Pharmaceutical sciences teaching	GastroPlus-based simulation teaching of biopharmaceutics successfully embedded in an undergraduate PharmD curriculum.
41	Obermeyer et al., 2019	Empirical study, USA	Governance, ethics, equity	A widely deployed commercial algorithm under-referred Black patients for care by using cost as a proxy for need.
42	Omiye et al., 2023	Empirical study, USA	Governance, ethics, equity	Four commercial large language models reproduced debunked race-based medical claims when queried clinically.

(*continued on next page*)

Table 3 (*continued*)

Ref.	Source	Setting	AI domain	Principal finding
45	Chalasani et al., 2023	Literature review	Clinical AI in pharmacy practice	Broad review documenting AI applications across pharmacy practice functions.
46	Wong et al., 2025	Forecast/ commentary, USA	Clinical decision support	Forecasts that AI-augmented decision support, documentation, and workflow automation will become routine within a decade.
47	Wang et al., 2025	Teaching innovation, China	Embedded course design	A teacher-AI-student interaction model embedded in existing case-based instruction strengthened applied clinical reasoning.
48	Luchen et al., 2024	Summit report, USA	Consortium strategy	ASHP Pharmacy Futures Summit convened consensus on consortium-based strategic directions for AI adoption.

Note. AI = artificial intelligence; LLM = large language model. Sources cited only in policy or accreditation documents (e.g., Health Canada, WHO, UNESCO, EU AI Act, AFPC, NAPRA, CCAPP, ACPE, GPhC, APC, PCNZ) are catalogued in Tables I and II rather than repeated here.

development, not curriculum design, is the first move (Section 9).

6.2. Interprofessional competencies: governance, ethics, equity, and regulatory awareness

Two competencies are specific to the health professions but not specific to pharmacy, and are best treated as shared interprofessional responsibilities.

AI governance, ethics, and equity. A substantial body of evidence documents bias in clinical AI. Obermeyer and colleagues[41] demonstrated that a widely deployed commercial algorithm used to triage high-risk patients for additional care substantially under referred Black patients because it used healthcare spending as a proxy for healthcare need. Omiye and colleagues[42] showed that four commercially available large language models, including those integrated into electronic health record systems, reproduced debunked, race-based medical claims about kidney function, lung capacity, pain thresholds, and skin thickness when queried in clinically relevant ways. WHO ethics and governance guidance articulates expectations of transparency, accountability, equity, and redress that graduates must be able to apply.[6,7] Graduates must be trained to apply sex and gender-based analysis plus (SGBA+) or equivalent frameworks to AI outputs, to recognize when a model's validation population does not match their patient population, and to consider Indigenous data sovereignty articulated in Canada through the OCAP® principles,[43] and equivalent data-governance frameworks elsewhere. The profession's ethical framework, articulated in Canada through the AFPC Professional role and NAPRA Principles of Professionalism and through counterpart codes internationally, must expand to encompass the harms AI can produce, amplify, and obscure. This competency is shared with medicine, nursing, dentistry, and allied health, and is most efficiently taught through interprofessional mechanisms.

The evolving regulatory environment. Health Canada's Pre-market Guidance for MLMDs introduced a formal framework for classification, predetermined change control plans, transparency, and post-market monitoring.[1] The European Union AI Act designates many health-related AI systems as high-risk and imposes obligations around data quality, transparency, and human oversight.[9] Sub-national statutes for example, Ontario's Strengthening Cyber Security and Building Trust in the Public Sector Act, 2024, and Quebec's health and social services information statute impose additional AI disclosure and oversight obligations, and Canada's Office of the Privacy Commissioner has proposed a regulatory framework for AI under a reformed federal privacy regime.[44] Pharmacy graduates will interpret these regulations for patients, employers, and colleagues, and should not meet them for the first time in their first job. Like governance and ethics, however, regulatory literacy

Table 4
A four-tier hierarchy of AI competencies for pharmacy graduates, mapped to AFPC pharmacist roles and example pedagogical strategies.

Tier	Competency	Primary AFPC role(s)	Example pedagogical strategy
General (upstream of pharmacy in steady state; transitional faculty responsibility)	Foundational AI literacy (knowledge)	Scholar	Short didactic modules introducing ML and LLM taxonomy; concept checks on training data, validation, drift, and hallucination.[37,38]
General	Critical appraisal of AI outputs (skill)	Scholar; Care Provider	Structured appraisal exercises using CLEAR or equivalent frameworks[39]; verification tasks flagging hallucinated citations.
General	Human–AI teaming and structured prompting (skill)	Care Provider; Collaborator	Prompt-engineering scaffolds in case-based learning; simulated-workflow exercises calibrating trust and override decisions.[25]
Interprofessional	AI governance, ethics, and equity	Professional; Health Advocate; Collaborator	Ethics seminars drawing on WHO AI-for-health guidance.[6,7]; SGBA+ and Indigenous-data-governance case analyses.
Interprofessional	Regulatory awareness	Professional; Leader–Manager	Regulatory-document analysis of GMLP principles, MLMD guidance, and the EU AI Act[1,9,15,16]; policy-interpretation workshops.
Pharmacy-specific	Clinical AI applications in pharmacy workflows	Care Provider; Collaborator	Simulated decision-support interpretation cases in medication reconciliation, dosing, counselling, and drug-information enquiry; human–AI OSCE stations.
Pharmacy-specific	Pharmaceutical sciences AI	Scholar; Care Provider	Integrated exercises in pharmaceutics (QSAR, AI-assisted formulation); ADMET-prediction interpretation laboratories; simulation-driven biopharmaceutics teaching.[40]
Pharmacy-specific	AI-augmented pharmacokinetics and model-informed precision dosing (the distinctive domain)	Care Provider	Bayesian-forecasting case exercises; MIPD interpretation assignments in clinical pharmacokinetics; integration of population-PK tools into clinical-pharmacokinetics assessment.[3]
Role-specific	Clinician/scientist/researcher specialization (introduction only at entry to practice)	All roles	Trajectory-specific electives and experiential placements; signposting to residency, graduate, and post-graduate pathways.

Note. AFPC = Association of Faculties of Pharmacy of Canada; ADMET = absorption, distribution, metabolism, excretion, and toxicity; CLEAR = Concise, Logical, Explicit, Adaptive, Reflective prompting framework; GMLP = Good Machine Learning Practice; LLM = large language model; ML = machine learning; MIPD = model-informed precision dosing; MLMD = machine-learning-enabled medical device; OSCE = Objective Structured Clinical Examination; QSAR = quantitative structure–activity relationship; SGBA+ = sex- and gender-based analysis plus.

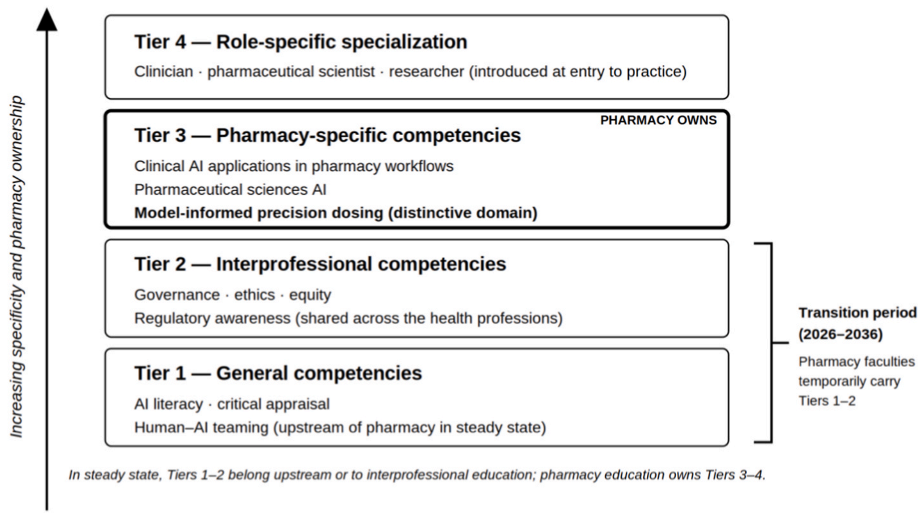

Fig. 1. The four-tier hierarchy of AI competencies for pharmacy graduates. In steady state, general (Tier 1) and interprofessional (Tier 2) competencies belong upstream of pharmacy education or to shared interprofessional instruction, while pharmacy education owns the pharmacy-specific (Tier 3) and role-specific (Tier 4) tiers, with model-informed precision dosing the most distinctive pharmacy owned domain. During the transition period (approximately 2026 to 2036), pharmacy faculties temporarily carry Tiers 1 and 2 because entering students have not received upstream AI preparation. AI = artificial intelligence.

is a shared health-profession responsibility; pharmacy specific application is a thin overlay on a broader competency that belongs at the interprofessional level.

6.3. *Pharmacy-specific competencies: where the profession has a distinctive claim*

Three competencies are specific to what pharmacists do, and represent the terrain on which pharmacy must make its case to the accreditor and to neighbouring professions. The three are not co-equal: AI-augmented pharmacokinetics and model-informed precision dosing (§6.3.3) carries the sharpest argument for a pharmacy-owned standard, while clinical AI in pharmacy workflows (§6.3.1) and pharmaceutical sciences AI (§6.3.2) are the contexts in which pharmacy's broader therapeutic and scientific expertise is exercised.

6.3.1. *Clinical AI applications in pharmacy workflows*

Pharmacists already work alongside AI-enabled clinical decision support systems, including drug-drug interaction flagging, renal dose adjustment prompts, AI-enhanced medication reconciliation, and, increasingly, large-language-model-generated patient counselling content.[2,45] A 2025 forecast prepared for the American College of Clinical Pharmacy anticipated that AI-augmented clinical decision support, documentation, and workflow automation would become routine across practice settings within the coming decade.[46] The pharmacy-specific layer of clinical AI competence is the ability to interrogate these outputs in the workflows where pharmacists are accountable: the medication-reconciliation conversation, the dosing intervention, the counselling encounter, the drug-information enquiry. Graduates should encounter these scenarios in pharmacy school rather than in first-job orientation. The general human-AI teaming skill of Section 6.1 is the substrate; the pharmacy-specific application is the habit of applying it in the places where pharmacists, and not other clinicians, carry the therapeutic decision.

6.3.2. *Pharmaceutical sciences AI*

The pharmaceutical sciences have been reshaped by AI at every stage of the drug development pipeline. Deep-learning approaches now generate de novo molecules, predict absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties, optimize excipient selection and formulation design, and accelerate physiologically-based pharmacokinetic modelling.[2] A pharmacy graduate who cannot read a quantitative structure-activity relationship paper, interpret a machine-learning ADMET prediction, or engage critically with an AI-assisted formulation proposal is working with half the vocabulary of contemporary pharmaceutical science. A Canadian exemplar already demonstrates that computationally augmented pharmaceutical sciences education is deliverable within existing course architectures: simulation driven biopharmaceutics teaching using GastroPlus® to visualize in vivo performance and pharmacokinetics for model drugs has been successfully embedded in the undergraduate PharmD curriculum at the University of Alberta Faculty of Pharmacy and Pharmaceutical Sciences,[40] providing a template other faculties can adapt rather than rebuild. This domain is not peripheral to the Care Provider role; it is foundational to the profession's claim to be the medication-therapy expert.

6.3.3. *AI-augmented pharmacokinetics and model-informed precision dosing*

Model-informed precision dosing (MIPD) is the pharmacy-proximal application of AI with the most mature evidence base, and it is where the profession's distinctive claim is strongest. Bayesian forecasting tools integrate population pharmacokinetic models with patient-specific data to produce individualized dose recommendations for vancomycin, aminoglycosides, tacrolimus, anticoagulants, and a growing list of other therapies.[3] Hybrid machine-learning and pharmacometric approaches are actively deployed. Population-PK Bayesian forecasting applied to the individual patient is more pharmacy than any other clinical discipline: it draws on pharmacokinetics, pharmacodynamics, therapeutics, and clinical pharmacology in the combination that defines pharmacy's intellectual centre of gravity. The absence of formal curricular treatment therefore risks a specific, avoidable harm ceding ground to commercial software used by other clinicians without critical pharmacotherapeutic oversight, and weakening the strongest single argument for pharmacy's continued authority over individualized medication decisions. For that reason, MIPD is treated here not as a subset of clinical AI or pharmaceutical sciences AI but as a competency domain in its own right. Every pharmacy graduate should leave school having interpreted, not merely observed, at least one MIPD case. Other AI applications, including pharmacovigilance and signal detection, medication adherence prediction, and population-health analytics, are also important to pharmacy practice; MIPD is singled out here because of its uniquely direct connection to the profession's traditional expertise in pharmacokinetics and individualized drug therapy, which is what makes its claim the least contested.

6.4. Role-specific specialization

Pharmacy graduates enter practice on three principal trajectories: the clinician, delivering medication therapy in hospital, community, or ambulatory settings; the pharmaceutical scientist, working in formulation, preclinical, or translational research; and the clinical or health-services researcher, conducting studies that are themselves increasingly AI-augmented. The competencies that matter most differ across these trajectories. MIPD and clinical decision support interpretation are most salient for clinicians; generative molecular design, QSAR interpretation, and ADMET prediction are most salient for pharmaceutical scientists; reproducibility, model reporting, and AI-assisted literature synthesis are most salient for researchers. A single entry-to-practice curriculum cannot deliver full role-specific mastery that is the work of residency, graduate training, and post-graduate specialization, but it should introduce all three trajectories sufficiently for graduates to choose among them and to read the literature of each.

7. Discussion: the timing problem, steady state versus transition

The hierarchy proposed in Section 6 describes a steady-state architecture: an end state in which pharmacy education owns only its pharmacy-specific and role-specific tiers. In this section we set out our own interpretation of how far pharmacy education currently is from that steady state, and what follows for sequencing decisions.

In steady state, the general tier (foundational AI literacy, critical appraisal, human AI teaming) belongs upstream of pharmacy education, and the interprofessional tier (governance, ethics, regulatory awareness) is shared across health-profession programmes. What pharmacy education owns, in steady state, is the pharmacy-specific and role-specific tiers described in Sections 6.3 and 6.4.

In our reading, that steady state does not yet exist, and the readiness findings reported in Section 5 are consistent with that view. Students entering pharmacy programmes in 2025-2027 did not encounter generative AI in K-12 curricula. Undergraduate pre-pharmacy science and statistics courses have not yet integrated machine-learning fundamentals consistently. Institution-wide digital-literacy supports for AI are uneven, and the faculty-student readiness gap documented in the national survey discussed in Section 5.[35] is, in our view, best read as a symptom of an unfinished upstream pipeline rather than a pharmacy-specific shortcoming. Waiting for that pipeline to fill before mandating AI competency in pharmacy would mean graduating a decade of pharmacists with unmet need.

Although the exact duration is uncertain and jurisdiction-dependent, we estimate that for the next five to ten years pharmacy faculties must carry a load of general and interprofessional AI content that structurally does not belong to them. We do not read this as a reason to refuse the load. Rather, it is our basis for recommending, in Sections 8 through 10, that programmes sequence faculty development first, curriculum second, and assessment third, and it is why we treat the AI capacity of the faculty delivering the curriculum, rather than curriculum design itself, as the binding constraint on AI-competent graduates (Section 9). It is also, in our assessment, the reason pharmacy should participate in, rather than wait for, the cross-disciplinary and cross-education-level conversations through which the steady state will eventually be built.

8. Discussion: implementation strategies for embedding AI in the pharmacy curriculum

The most common objection to curricular additions of this kind is that the pharmacy degree is already full. It is. Building on the pedagogical evidence reported in Section 5, we do not recommend a standalone AI course, which would be dated before students completed it, but instead propose the integration of AI competencies across the existing curriculum. This recommendation follows directly from the findings above: the 2026 analysis of educational strategies at a US college of pharmacy reached the same conclusion, and the national faculty-student survey and five-school simulated-patient collaboration summarized in Section 5 illustrate that integration into existing coursework, rather than a separate subject, is both feasible and well received.[34–36] We propose five implementation principles below, summarized alongside the faculty-development principle treated separately in Section 9, in Table V.

First, we recommend that programmes integrate, not silo: AI competencies should appear in pharmacology (pharmacogenomic prediction, drug-drug interaction prediction), pharmaceutics (quantitative structure-activity relationships, AI-assisted formulation), pharmacokinetics (Bayesian forecasting, MIPD), therapeutics (clinical decision support interpretation), professional practice (ethics, equity, governance), and experiential learning (structured reflection on the AI tools students encounter on rotations).[2,3] A standalone "AI in pharmacy" course is, in our view, the worst of both worlds: it consumes curricular time while failing to embed competency where it will be used. A recent teaching innovation embedded within a required pharmacy course, combining structured questioning-based training with a teacher-AI-student interaction model directly inside existing case-based

Table 5

Pedagogical implementation principles for embedding AI competencies across the pharmacy curriculum. Faculty development is set apart from the other principles because the other five depend on it (see Section 9).

Principle	Rationale	Practical curricular implication
Faculty development first (see Section 9)	Faculty readiness is the binding constraint; the remaining principles cannot be delivered without it.	A documented, resourced, train-the-trainer faculty-development plan precedes curriculum mapping and assessment design, with a three- to five-year runway.
Integrate, do not silo	Competencies developed in isolation are not transferred to the clinical contexts in which AI is actually used.	AI content is distributed across pharmacology, pharmaceutics, pharmacokinetics, therapeutics, professional practice, and experiential learning, rather than housed in a stand-alone course.
Teach failure modes before capability modes	First impressions bias subsequent appraisal; students who meet generative AI as a shortcut may not later apply the critical frame.	Hallucination, algorithmic bias, alert fatigue, automation bias, and distributional drift are taught before chatbot-assisted case work.
Assess with real cases	Assessment shapes curriculum more reliably than policy.	OSCE stations include recognition of AI hallucination, challenge to an inappropriate decision-support recommendation, and critique of an automated dose suggestion that conflicts with evidence.
Build national and international infrastructure interprofessionally	Eleven faculties reinventing the same curriculum independently is inefficient and inconsistent.	Shared, modular, version-controlled content is developed through national consortia drawing on pharmacy, computer science, data science, and ethics.
Update informatics competencies with explicit AI content	Existing informatics competency frameworks predate generative AI and the MLMD regulatory environment.	An explicit AI extension is added to the AFPC–Infoway indicators (and equivalents elsewhere) addressing ML, LLMs, algorithmic bias, model governance, and MLMD regulation.

Note. AFPC = Association of Faculties of Pharmacy of Canada; LLM = large language model; ML = machine learning; MLMD = machine-learning-enabled medical device; OSCE = Objective Structured Clinical Examination.

instruction, demonstrated that this kind of embedded approach can strengthen applied clinical reasoning without adding a separate course to an already crowded timetable.[47]

Second, we recommend teaching failure modes before capability modes: hallucination, algorithmic bias, alert fatigue, automation bias, distributional drift, and adversarial examples should appear in the curriculum before chatbot-assisted case work. Students need the critical frame before they are given the tool.[41,42] A student who is first introduced to generative AI as a useful drug-information shortcut, and only later learns it confabulates citations and reproduces race-based medical myths, will carry the first impression further than the second correction.

Third, we recommend assessing with real cases: Objective Structured Clinical Examination (OSCE) stations can and should include scenarios in which students must recognize an AI hallucination, identify an inappropriate decision-support recommendation, or challenge an automated dose suggestion that conflicts with available clinical evidence. Assessment shapes curriculum more reliably than policy statements; where accreditors require evidence of AI competency assessment, programs produce it.

Fourth, we recommend building national and international infrastructure interprofessionally: the AFPC-Canada Health Infoway collaboration on informatics is one model.[20] Internationally, FIP's educational infrastructure provides a comparable vehicle.[5,12] National AI in pharmacy educational consortia drawing on faculties of pharmacy, departments of computer science, data science, and ethics, and national research councils can develop modular, shared, version-controlled content that each faculty adopts and adapts, rather than reinventing the same curriculum independently at every institution. ASHP's 2024 Pharmacy Futures Summit on Artificial Intelligence in Pharmacy Practice reached a similar consensus, convening practitioners and thought leaders to chart consortium-based strategic directions for AI adoption across pharmacy practice, a model that pharmacy education could adapt for curricular content development.[48]

Fifth, we recommend updating informatics competencies with explicit AI content: the thirty-one AFPC Infoway indicators remain a sound foundation but require an explicit AI extension addressing machine learning, large language models, algorithmic bias, model governance, and the MLMD regulatory environment.[20] Analogous informatics competency frameworks in other jurisdictions need corresponding updates.

9. Discussion: faculty development as the binding constraint

Of the pedagogical principles summarized above, faculty development is not one among several; it is the principle on which the others depend, and for that reason we treat it here as a standalone section rather than a sixth bullet. A curriculum that maps AI competencies across pharmacology, pharmaceutics, pharmacokinetics, and therapeutics cannot be delivered by faculty who have not themselves acquired those competencies; an OSCE station that probes a student's recognition of AI hallucination cannot be assessed by an examiner who does not recognize hallucination when they see it. Published analyses consistently identify faculty readiness as the binding constraint.[11,21,25]

The faculty-development challenge is larger than is commonly acknowledged because health-profession educators are routinely asked to develop expertise in two professional domains at once: the disciplinary domain they were trained in (clinical pharmacy, pharmaceutical sciences, or a pharmacy-adjacent basic science) and the pedagogical domain (evidence-based teaching, assessment design, curricular alignment) typically without formal preparation in the second. For clinician educators and researcher educators, the teaching role is often a third professional identity layered on top of clinical or research practice. AI content adds a further domain to that load: technical literacy, facility with generative tools, and instructor-ready critical-appraisal skill. Expecting faculty to acquire this while continuing to practise, publish, or both, and while being assessed, in accreditation, on student outcomes is a significant ask. It is nevertheless the ask the profession faces, and pretending otherwise is, in our view, how the transitional load discussed in Section 7 becomes a permanent drag on the curriculum.

Accreditation is the mechanism by which this ask becomes a requirement rather than an aspiration. CCAPP Standard 20 (faculty and staff training and professional development) and its international counterparts (ACPE standards addressing faculty and preceptor adequacy, the GPhC's requirements for educator capability, and analogous APC and PCNZ provisions) are the single highest-leverage accreditation instruments available for this purpose. A standard that requires programmes to demonstrate, as a condition of accreditation, a documented and resourced AI faculty-development plan with named outcomes, timelines, and measurable indicators of faculty AI capability puts the responsibility for sequencing faculty first on the institution rather than on the committed individual instructor. Such indicators can be concrete and auditable: the proportion of faculty completing structured AI training, the number of AI-integrated courses and AI-focused OSCE stations, faculty AI confidence scores on a validated self-assessment instrument, and preceptor AI competency completion rates. Standard 20 is where the AI transition lives or fails. An accreditor that writes faculty development into Standard 20 by 2027 will see its graduates enter AI-augmented practice in 2031 prepared for it; an accreditor that does not will graduate cohorts whose faculty have not been systematically equipped to teach the material, and which the profession will have to retrain after licensure.

A realistic faculty development architecture comprises three elements: (a) a train the trainer core, in which a small group of AI fluent faculty are resourced to teach colleagues rather than to reinvent content independently; (b) protected time, recognized in workload calculations, for faculty to acquire and maintain AI competence on the understanding that this is professional development, not personal enrichment; and (c) cross institutional sharing of content, assessment, and exemplar materials through the national consortia described in Section 8. Absent any one of these, the standard becomes a compliance exercise rather than a capability build.

10. Discussion: accreditation and policy levers

A competency framework not reflected in accreditation is aspirational. Accreditation is the mechanism by which educational outcomes become requirements, and by which faculties allocate curricular time and faculty effort. The reviewed literature and accreditation documents support five concrete levers that accrediting bodies internationally can apply (Table VI). We present them here in order of near-term leverage, with the faculty-development lever placed first for the reasons set out in Section 9.

First, we recommend that standards addressing faculty development, CCAPP Standard 20 and its international counterparts, require a documented and resourced AI faculty-development plan as a condition of accreditation. Faculty capacity is the binding constraint: a student cannot be taught what a faculty member does not know, and preceptor capacity is at least as constrained as faculty capacity.[11] This is the lever on which the other four depend.

Second, accreditors can incorporate an explicit AI competency domain into their educational outcomes framework on the structural footing of other recent curricular inclusions. In Canada, the precedent of Indigenous health integration advanced in response to the Truth and Reconciliation Commission's Calls to Action[33] is directly relevant: the profession recognized a major curricular gap of moral and practical consequence and mandated its inclusion through accreditation. Structurally analogous precedents exist in Australia, where Aboriginal and Torres Strait Islander cultural safety is embedded across the APC standards,[30] and in New Zealand, where Te Tiriti o Waitangi constitutes a dedicated domain in the competence standards.[32] In each instance the profession recognized a curricular gap of consequence and mandated its inclusion through accreditation architecture. AI competence is another

Table 6
Five accreditation and policy levers available to pharmacy accreditors internationally for embedding AI competence. Levers are presented in order of near-term leverage; faculty development is placed first because the remaining four depend on it.

Lever	Mechanism	Illustrative standards or precedents
Mandate a documented, resourced AI faculty-development plan (highest-leverage)	Require programmes to demonstrate AI capacity among faculty and preceptors as a condition of accreditation, with named outcomes, timelines, and indicators.	CCAPP Standard 20 and its international counterparts.
Incorporate an explicit AI competency domain	Add AI competence to the educational outcomes framework on the structural footing of other recent transversal competencies, attributing competencies to the hierarchy tier that owns them.	Indigenous health integration in Canada via Truth and Reconciliation Commission Calls to Action[33]; Aboriginal and Torres Strait Islander cultural safety[30]; Te Tiriti o Waitangi.[32]
Require assessable evidence of AI competency	Demand OSCE stations, capstone assessments, or specific APPE objectives rather than elective lecture content alone.	CCAPP Standard 1 (Educational Outcomes) and Standard 6 (Teaching, Learning, and Assessment)[19]; analogous ACPE and GPhC standards.[23,24]
Require an institutional AI-use policy	Distinguish prohibited, permitted, and required uses of AI tools in coursework, assessment, and practice experience.	UNESCO guidance on generative AI in education and research.[8]
Align with federal, regional, and international frameworks	Require teaching of the regulatory architecture that governs the AI tools graduates will use.	GMLP guiding principles[15]; Transparency guiding principles[16]; Health Canada[1]; EU AI Act.[9]

Note. ACPE = Accreditation Council for Pharmacy Education; APC = Australian Pharmacy Council; APPE = advanced pharmacy practice experience; CCAPP = Canadian Council for Accreditation of Pharmacy Programs; GMLP = Good Machine Learning Practice; GPhC = General Pharmaceutical Council; OSCE = Objective Structured Clinical Examination; PCNZ = Pharmacy Council of New Zealand; TRC = Truth and Reconciliation Commission of Canada.

such gap; the fact that it is technologically rather than socially rooted does not make it an obligation the profession can defer. The hierarchy proposed in Section 6 informs how we believe this lever should be used: accreditors should mandate pharmacy specific and role specific competencies directly, and require programmes to evidence interprofessional and general competencies with appropriate attribution to upstream or shared instruction.

Third, accreditors can require programmes to demonstrate that students have achieved AI competencies through assessable evidence: OSCE stations, capstone assessments, or specific objectives in advanced pharmacy practice experiences rather than through elective lecture content alone. In the Canadian context this implies revisions to CCAPP Standard 1 (Educational Outcomes) and Standard 6 (Teaching, Learning, and Assessment)[19]; analogous standards exist in the ACPE and GPhC frameworks.[23,24]

Fourth, programs can be required to hold an institutional policy on AI use in coursework, assessment, and practice experience, analogous to existing academic-integrity policies but explicit about the distinction between prohibited, permitted, and required uses of AI tools. UNESCO guidance on generative AI in education and research is aligned with this approach.[8] Programs should teach the professional use of AI tools rather than merely police their misuse.

Fifth, standards can be aligned with federal, regional, and international frameworks. Programs should teach the Good Machine Learning Practice guiding principles,[15] the Transparency for Machine-Learning-Enabled Medical Devices guiding principles,[16] relevant national MLMD guidance,[1] and supranational instruments such as the EU AI Act.[9] These documents are not optional enrichment; they are the regulatory architecture within which graduates will practice.

A cross-cutting consideration shapes how we believe all five levers should be written: the pace of change. AI tools evolve faster than accreditation revision cycles, so specific platforms, models, and regulatory frameworks will date more quickly than the curriculum that references them. For that reason we recommend that AI competencies be revisited every two to three years, and that standards anchor on enduring competencies, namely critical appraisal, governance, and human-AI teaming, rather than on mastery of any individual tool. Writing standards around durable capabilities rather than named products is, in our assessment, what allows an accreditation framework to remain current between formal revision cycles.

11. General discussion

Pharmacy has argued successfully, over more than twenty years, that pharmacists are the medication-therapy experts. That argument has underpinned expanded scope of practice, pharmacist prescribing authority in many jurisdictions, and the profession's entrenchment in primary care. AI now intermediates the exercise of that expertise. A clinical recommendation in 2030 will not be generated by a pharmacist alone; it will be generated by a pharmacist working alongside an AI enabled clinical decision support system, a generative tool that drafts patient counselling, an MIPD platform that individualizes dosing, and a pharmacovigilance signal detection algorithm that flags potential adverse events.[2,3,45] The question that will determine the profession's next twenty years is who owns the expertise to use those tools well?

The hierarchy proposed here is our attempt to answer that question with a policy useful level of specificity. Foundational AI literacy, critical appraisal of AI outputs, and human AI teaming are not pharmacy competencies in steady state; they are general professional skills that belong upstream. Governance, ethics, and regulatory awareness are shared health profession competencies and are most efficiently taught interprofessionally. What is distinctively pharmacy's, and therefore what accreditation must own, is clinical AI applied in pharmacy workflows, pharmaceutical sciences AI, and most pointedly AI augmented pharmacokinetics and model-informed precision dosing. The accreditor that mandates faculty capacity in these areas, and assessable student evidence within them, will graduate clinicians most literate in model informed pharmacotherapy. If pharmacy graduates are those clinicians, the case for pharmacist authority extends naturally into this new layer of practice. If they are not, another profession will make the case and win it.

The argument implies a broader conversation that is beyond the scope of this review but that we believe the pharmacy academy should participate in shaping rather than wait to receive. Working out which AI competencies are owned by which level of the education system, and by which discipline within health professions education, is a cross disciplinary and cross education level agenda spanning K-12 systems, undergraduate science, institutional digital literacy infrastructure, and the post-licensure continuing-education pathways by which currently practising pharmacists will acquire what their degree did not contain. Pharmacy's distinctive contribution to that broader conversation, and the argument we have tried to make here, is that MIPD and AI augmented pharmacotherapy are areas in which the profession has an unambiguous and currently under claimed stake.

A focus on entry-to-practice education should not obscure a parallel need. Most pharmacists who will practise in the AI-augmented environment of 2030 have already graduated, so continuing professional development will be as consequential as curricular reform, and professional regulators may in time require AI-related continuing education. The hierarchy proposed here can structure that provision as readily as it structures the entry-to-practice curriculum: practising pharmacists most need the pharmacy-specific tier, and model-informed precision dosing in particular, layered onto the general and interprofessional competencies their initial training did not provide.

To our knowledge, this is the first international pharmacy education review to propose a competency hierarchy that explicitly distinguishes general, interprofessional, pharmacy-specific, and role-specific AI competencies and links each tier to the accreditation mechanism responsible for it.

This narrative review has limitations. It is neither systematic nor registered; it draws primarily on English-language literature and institutional documents; and it reflects the regulatory landscape as of early 2026 in a domain moving at generational speed. Its purpose is synthetic and pragmatic: to place pharmacy education's AI related challenges in an international frame and to describe the competency, pedagogical, and accreditation levers that the literature supports. Future work should include formal scoping and systematic reviews of AI curricular interventions in pharmacy, empirical evaluations of the implementation principles described here, longitudinal studies of student and faculty competency development, and cross jurisdictional comparative studies of accreditation standards as they evolve. Table VII sets out priority directions for that work.

12. Conclusion

The deprescribing movement argued that a discipline should be built around undoing pharmacotherapeutic decisions that should have been made correctly in the first place. The AI transition is its mirror image: an opportunity to make pharmacotherapeutic decisions better from the outset: better informed, better individualized, and better monitored provided that the profession that claims the expertise actually learns to use the new tools. Accreditation standards, working in concert with faculty development, integrated curriculum design, and rigorous assessment, are the instruments by which that learning is guaranteed. CCAPP Standard 20 and its international counterparts are where the work begins. They should be revised accordingly, and soon.

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Neal M. Davies: Conceptualization, Data curation, Formal analysis, Investigation, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kamal Dua:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Raimar Löbenberg:** Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing.

Table 7
Priority directions for future research on artificial intelligence in pharmacy education.

Priority direction	Why it matters
Validated AI competency assessment tools	No pharmacy-specific, psychometrically validated instruments yet exist, though embedded teaching-innovation models that pair structured assessment with AI-supported instruction offer an early template for what a validated instrument could build upon.[47]
Faculty AI readiness surveys	Faculty capacity is the principal implementation barrier and is only beginning to be measured: a national survey has established an early baseline of faculty and student AI familiarity, trust, and training,[35] and a five-programme implementation study has begun to document the practical and institutional barriers faculty encounter when embedding AI tools into existing skills instruction.[36]
AI-focused OSCE development	Assessable evidence is needed to support accreditation requirements.
MIPD educational outcomes	The distinctive pharmacy-owned domain, with outcomes that remain undefined.
Longitudinal graduate follow-up	Required to determine the real-world practice impact of curricular change.

Note. AI = artificial intelligence; MIPD = model-informed precision dosing; OSCE = Objective Structured Clinical Examination.

Conflicts of interest

None declared.

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